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PROVIDED BY :- ER OMKAR TULE

SOLVED MODEL QUESTION PAPER 1 - AS PER 2024 PATTERN

Engineering Chemistry [BSC-103-BES]

F.E. (Common to All Branches) Sem I/II (End Sem)

Time : 2 $\frac{1}{2}$ Hours] [Maximum Marks : 70

Instructions to the candidates :

1) *Q. No. 1 is compulsory. Solve Q. No. 2 or Q. No. 3, Q. No. 4 or Q. No. 5, Q. No. 6 or Q. No. 7, Q. No. 8 or Q. No. 9, Q. No. 10 or Q. No. 11.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicates full marks.*
4) *Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.*
5) *Assume suitable data, if necessary.*

Q.1 Multiple Choice Questions (1 Mark each) [10]

a) *Permanent hardness in water is caused by _____ impurity.*

a) $Ca(HCO_3)_2$ *b)* KCl *c)* $MgSO_4$ *d)* Na_2CO_3

Ans. : c) $MgSO_4$

b) *In zeolite process exhausted zeolite is regenerated by using _____.*

a) $NaOH$ *b)* $NaCl$ *c)* HCl *d)* $ZnCl_2$

Ans. : b) NaCl

c) *Glass electrode is used as _____ in pH meter.*

a) primary reference electrode b) Ion selective electrode c) indicator electrode d) both b and c

Ans. : d) both b and c

d) *In UV - visible spectroscopy, when absorption maxima shifts towards longer wavelength, it is called as _____.*

a) bathochromic shift b) hypsochromic shift c) hyperchromic shift d) hypochromic shift

Ans. : a) bathochromic shift

e) *_____ is a example of biosynthetic polymer.*

a) Kevlar b) Polyacetylene c) PHBV d) Nylon 6

Ans. : c) PHBV

f) *Fullerenes are _____ type of nanomaterials.*

a) 0-D b) 1-D c) 2-D d) 3-D

Ans. : a) 0-D

g) *In proximate analysis of coal, crucible with powdered coal is heated at _____ temperature for _____ minutes while determining moisture in it.*

a) 90°, 30 *b)* 90°, 60 *c)* 105°, 60 *d)* 105°, 30

Ans. : c) 105°, 60

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h) *Biodiesel is obtained from vegetable oil by process called _____.*

a) oxidation b) esterification c) hydrogenation d) transesterification

Ans. : d) transesterification

I) *In cathodic protection method, sacrificial anode used is made up of _____ metal.*

a) Copper b) Manganese c) Zinc d) Sodium

Ans. : c) Zinc

J) *When base metal like iron is coated with copper it is a type of _____ coating.*

a) anodic coating b) cathodic coating c) non - metallic coating d) powder coating

Ans. : b) cathodic coating

Q.2 a) *i) Justify the use of EBT indicator in determination of hardness of water by EDTA method.* **(Refer section 1.4)**

ii) 50 ml of water sample consumed 11.5 ml of 0.04 M EDTA whereas 50 ml of boiled water sample consumed 7.5 ml of same EDTA. Calculate temporary, permanent and total hardness of water sample. **[3 + 3]**

Ans. : Temporary hardness = 320 ppm, Permanent hardness = 600 ppm, Total hardness = 920 ppm.

b) *A sample of tanker water contains $MgSO_4$, $Mg(HCO_3)_2$ and $CaCl_2$ salts. Write the chemical reactions involved in the cation and anion exchanger during softening of such water by Ion exchange process.* **(Refer section 1.7.3)** **[3]**

c) *Explain working of atmospheric water generator used in trapping of water from air.* **(Refer section 1.9)** **[3]**

OR

Q.3 a) *i) A sample of tanker water contains $CaSO_4$, $Mg(NO_3)_2$ and $MgCl_2$ salts. Write the chemical reactions involved in zeolite process during softening of such water.* **(Refer section 1.7.1)**

ii) Zeolite softener was completely exhausted and was regenerated by passing 150 lit. of NaCl solution containing 50 gm NaCl per lit. How many lit. of sample water of hardness 200 ppm can be softened by this softener ? **[3 + 3]**

Ans. : Sample water softened = 32051.28 lit.

b) *Water sample is alkaline to both phenolphthalein and methyl orange indicator., 50 ml of water sample on titration with N/50 HCl required 7.2 ml for phenolphthalein end point and 18.5 ml for complete neutralization. Determine the type and amount of alkalinity present in water.* **[3]**

Ans. : CO_3^{-2} alkalinity = 288 ppm and HCO_3^- alkalinity = 20 ppm.

c) *Compare : Sludge with scales.* **(Refer section 1.6)** **[3]**

Q.4 a) *Draw block diagram of double beam spectrophotometer and explain function of each component.* **(Refer section 2.19)** **[6]**

b) *Explain construction of glass electrode with diagram and state its advantages.* **(Refer section 2.4.1)** **[3]**

c) *Define following terms :*

1. Hypsochromic shift 2. Hypochromic shift 3. Auxochrome **(Refer section 2.18)** **[3]**

OR

Q.5 a) *Explain the various stages with the help of titration curve in conductometric titration of strong acid (HCl) against strong base (NaOH) and give the reaction involved in the titration.* **(Refer section 2.8)** **[6]**

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b) *Calculate the molar absorptivity of 5×10^{-4} M solution which has an absorbance of 0.45 and pathlength 2 cm.* [3]

Ans. : Molar absorptivity = 0.045×10^{4} $dm^3/mol/cm$.

c) *Give the composition of the membrane used in ion selective electrode used in the estimation of i) H^+ ions, ii) Cl^- ions, iii) F^- ions.* **(Refer section 2.5.1)** [3]

Q.6 a) *What are biodegradable polymers ? Give important features of biodegradable polymer. How are they classified ? Give suitable example for each type.* **(Refer section 3.5.3)** [6]

b) *Give preparation of polycarbonate using bisphenol A and phosgene. Give two applications of polycarbonate.* **(Refer section 3.5.2)** [3]

c) *Explain how quantum dots show change in colour with change in size.* **(Refer section 3.11.3)** [3]

OR

Q.7 a) *Give preparation, properties and application of 1. PHBV, 2. Polycarbonate.* **(Refer sections 3.5.2 and 3.5.3)** [6]

b) *Give structure of cis and trans polyacetylene. How cis polyacetylene is converted into trans polyacetylene ?* **(Refer section 3.5.4.6)** [3]

c) *Explain structure of graphene with figure and give two applications of graphene.* **(Refer section 3.11.1)** [3]

Q.8 a) *Give principle of Bomb calorimeter. Calculate GCV and NCV of fuel containing 3 % hydrogen if 0.65 gm. of fuel on complete combustion in Bomb calorimeter increased the temperature of water in the calorimeter from 21.5 °C to 24.1 °C . The mass of water in the calorimeter was found to be 2000 gm. and water equivalent of calorimeter is 625 gm.* [6]

Ans. : GCV = 10500 cal/gm and NCV = 10341.51 cal/gm.

b) *Give principle, procedure and formula for determination of % sulphur in ultimate analysis of coal.* **(Refer section 4.5.3)** [3]

c) *What is biodiesel ? Give its preparation and state two advantages.* **(Refer section 4.6.2)** [3]

OR

Q.9 a) *1.86 gm. of coal sample after heating in silica crucible at 110 °C for 1 hour weighed 1.75 gm. The crucible then covered with a vented lid and strongly heated further at 950 °C for exactly 7 min. The coal on cooling weighed 1.63 gm. The crucible is further heated without a lid at 725 °C for 30 min. The residue weighed 0.22 gm. Calculate Proximate analysis result. Evaluate the above coal sample based on proximate analysis result.* [6]

Ans. : Moisture % = 5.91, Volatile matter % 6.45, Ash % = 11.83, FC % = 75.81.

b) *Define GCV and NCV. Explain the relation between them.* **(Refer sections 4.2.2, 4.2.3 and 4.2.4)** [3]

c) *Explain the preparation of hydrogen by steam reforming of methane with reactions involved.* **(Refer section 4.7)** [3]

Q.10 a) *What are the different conditions under which wet corrosion occurs ? Explain with suitable example oxygen absorption mechanism of wet corrosion.* **(Refer section 5.3)** [6]

b) *Explain any three factors influencing rate of corrosion with respect to nature of metal.* **(Refer section 5.4.1)** [3]

c) *Compare : Anodic coating and Cathodic coating.* **(Refer section 5.5.3)** [3]

OR

Q.11 a) *Explain cathodic protection method using impressed current with respect to principle, method and application.* **(Refer section 5.5.1)** [6]

b) *Define oxidation corrosion. What are the different types of metal oxides formed ?* **(Refer section 5.2.1)** [3]

c) *Discuss the use of metal nanoparticles in paints.* **(Refer section 5.5.4)** [3]

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SOLVED MODEL QUESTION PAPER 2 - AS PER 2024 PATTERN

Engineering Chemistry [BSC-103-BES]

F.E. (Common to All Branches) Sem I/II (End Sem)

Time : 2 $\frac{1}{2}$ Hours] [Maximum Marks : 70

Q.1 Multiple Choice Questions (1 Mark each) [10]

a) *The loose and slimy precipitate formed during continuous boiling of water in the boiler is called as _____.*

a) Sludge *b) Scale* *c) Coating* *d) Layer*

Ans. : a) Sludge

b) *In Ion exchange process for water purification, exhausted cation exchangers are regenerated by using _____.*

a) NaOH *b) NaCl* *c) HCl* *d) $ZnCl_2$*

Ans. : c) HCl

c) *Calomel electrode is an example of _____.*

a) Primary reference electrode *b) Secondary reference electrode*

c) Indicator electrode *d) Electrode with no fixed potential*

Ans. : b) Secondary reference electrode

d) *In UV - visible spectroscopy, when absorption maxima shifts towards shorter wavelength, it is called as _____.*

a) Bathochromic shift *b) Hypsochromic shift*

c) Hyperchromic shift *d) Hypochromic shift*

Ans. : b) Hypsochromic shift

e) *N-doping in conducting polymer is done by _____.*

a) I_2 *b) $FeCl_3$* *c) Na* *d) Br_2*

Ans. : c) Na

f) *Nanowires are _____ nanomaterials.*

a) 0-D *b) 1-D* *c) 2-D* *d) 3-D*

Ans. : b) 1-D

g) *Carbon dioxide formed in steam reforming process of manufacturing hydrogen is separated by _____.*

a) Compression and cooling *b) Reaction with K_2CO_3*

c) Either of (a) or (b) *d) None of these*

Ans. : c) Either of (a) or (b)

h) *In power alcohol, _____ ethanol is blended with petrol and used as fuel in IC engine.*

a) 80 - 90 % *b) 40 - 50 %*

c) 20 - 25 % *d) less than 10 %*

Ans. : c) 20 - 25 %

i) *Rusting of Iron in acid occurs with evolution of _____.*

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a) Nitrogen b) Oxygen
c) Chlorine d) Hydrogen

Ans. : d) Hydrogen

j) *Tinning is an example of* _____.

a) Anodic coating b) Cathodic coating
c) Non metallic coating d) Powder coating

Ans. : b) Cathodic coating

Q.2 a) *i) Justify the use of buffer with pH = 10 in determination of hardness of water by EDTA method.* **(Refer section 1.4)**

ii) 25 ml of water sample consumed 18.5 ml of 0.02 M EDTA where as 25 ml of boiled water sample consumed 11.5 ml of same EDTA. Calculate temporary, permanent and total hardness of water sample. **[6]**

Ans. : Temporary hardness = 560 ppm, Permanent hardness = 920 ppm, Total hardness = 1480 ppm.

b) *Give causes, disadvantages and preventive measures for foaming in boilers.* **(Refer section 1.6)** **[3]**

c) *Explain reverse osmosis process with figure, principle and process.* **(Refer section 1.8.1)** **[3]**

OR

Q.3 a) *What is Priming in boiler? Differentiate scales and sludges in boiler. (Give 5 points of differences).* **(Refer section 1.6)** **[6]**

b) *Water sample is not alkaline to phenolphthalein indicator. However, 100 ml of water sample on titration with N/50 H_2SO_4 required 16.4 ml for methyl orange end point. Determine the type and amount of alkalinity present in water.*

Ans. : HCO_3^- alkalinity = 164 ppm **[3]**

c) *What are zeolites ? Give principle of water softening using zeolite. Explain how exhausted zeolites are regenerated ?.* **(Refer section 1.7.1)** **[3]**

Q.4 a) *Explain different types of electronic transitions that occur in an organic molecule after absorbing UV radiations.* **(Refer section 2.17)** **[6]**

b) *Explain construction of conductivity cell with labelled diagram.* **(Refer section 2.7)** **[3]**

c) *What are the reference and indicator electrodes used for measurement of pH ? Draw figure and give representation of indicator electrode used for measurement of pH.* **(Refer sections 2.4.1 and 2.10.1)** **[3]**

OR

Q.5 a) *Give the procedure for standardisation of pH meter and the procedure for titration of strong acid verses strong base using pH meter. Draw titration curve and give the reaction involved.* **(Refer sections 2.10.2 and 2.11)** **[6]**

b) *State Lambert's law and Beer's law. Give the formula to calculate molar extinction coefficient.* **(Refer section 2.16)** **[3]**

c) *Discuss composition and working of Fluoride ion selective electrode with neat labelled diagram.* **(Refer section 2.5.1)** **[3]**

Q.6 a) *What are Carbon nanotubes? Give their types and any 4 applications.* **(Refer section 3.11.2)** **[6]**

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b) *Define functionality of monomers. Explain bifunctional and trifunctional monomers with examples.* **(Refer section 3.3)** **[3]**

c) *What are Quantum dots? Give 2 properties and 2 applications of quantum dots.* **(Refer section 3.11.3)** **[3]**

OR

Q.7 a) *Explain with diagram the structure of Graphene. Give 3 properties and 3 applications of Graphene.* **(Refer section 3.11.1)** **[6]**

b) *Explain 3 factors responsible for biodegradation of polymers.* **(Refer section 3.5.3.4)** **[3]**

c) *Differentiate between Thermoplastic and Thermosetting polymers. (Give 3 points of differences)* **(Refer section 3.4)** **[3]**

Q.8 a) *What is Proximate analysis of coal ? Mention the principle involved in the analysis of each constituent. Give the significance of any one constituent.* **(Refer section 4.5.1)** **[6]**

b) *Classify fuel on the basis of chemical reactions with examples.* **(Refer section 4.1.2)** **[3]**

c) *0.2 gm coal sample on complete combustion in presence of continuous oxygen supply, increased the weight of U-tube containing $CaCl_2$ by 0.10 gm and U-tube containing KOH by 0.52 gm. Calculate % C and % H present in the coal.* **[3]**

Ans. : % C = 70.90 % , % H = 5.55 %

OR

Q.9 a) *Give principle of Boy's gas calorimeter. In a Boy's gas calorimeter, a gaseous fuel was burnt and the following data was obtained :*

Volume of gas burnt = 0.1 m^3 at STP

Weight of water heated = 10 kg

Temperature of inlet water = 22 °C

Temperature of outlet water = 34 °C

Weight of steam condensed = 0.025 kg

Calculate the gross and net calorific values of the fuel. Assume the latent heat of water vapours formed as 587 kcal / kg. **[6]**

Ans. : GCV = 1200 kcal/ m^3 , NCV = 1053.25 kcal/ m^3

b) *Give 3 advantages and 3 disadvantages of power alcohol.* **(Refer section 4.6.1)** **[3]**

c) *Explain the construction of Lithium Ion battery with figure.* **(Refer sections 4.8 and 4.8.1)** **[3]**

Q.10 a) *Explain Hydrogen evolution and oxygen absorption mechanism of wet corrosion.* **(Refer section 5.3.2)** **[6]**

b) *Give oxidation reaction involved and state the type of oxide film formed on the surface of Na, Ag and Cr metals.* **(Refer sections 5.2.1)** **[3]**

c) *Explain anodic coating and cathodic coating. Which is better ?* **(Refer section 5.5.3)** **[3]**

OR

Q.11 a) *Discuss any six factors affecting the rate of corrosion.* **(Refer sections 5.4.1 and 5.4.2)** **[6]**

b) *Explain impressed current cathodic protection of metal.* **(Refer sections 5.5.1)** **[3]**

c) *What is Pilling Bedworth rule ? Give its significance along with examples.* **(Refer section 5.2.2)** **[3]**

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SOLVED MODEL QUESTION PAPER 3 - AS PER 2024 PATTERN

Engineering Chemistry [BSC-103-BES]

F.E. (Common to All Branches) Sem I/II (End Sem)

Time : 2 $\frac{1}{2}$ Hours] [Maximum Marks : 70

Q.1 Multiple Choice Questions (1 Mark each) [10]

i) *Temporary hardness is also known* _____.

a) Carbonate hardness *b) Non-carbonate hardness*

c) Total hardness *d) None of above*

Ans. : a) Carbonate hardness

ii) *Alkalinity of water is due to* _____.

a) OH^- *b)* CO_3^{-2}

c) HCO_3^- *d) All of these*

Ans. : d) All of these

iii) *Name the electrolyte used in solid state ion selective electrode, used for determination of fluoride ion* _____.

a) BF_3 *b) NaF* *c) LiCl* *d)* LaF_3

Ans. : b) NaF

iv) *Spectrophotometer working is based on* _____.

a) Stephan Boltzmann law *b) Coulomb law*

c) Faraday *d) Beer Lambert law*

Ans. : d) Beer Lambert law

v) *During n-doping of polyacetylene* _____ *occurs.*

a) reduction *b) hydrolysis*

c) oxidation *d) degradation*

Ans. : a) reduction

vi) *Nanomaterials are the materials in which size of particles ranges from* _____.

a) 1 nm - 100 nm *b) 1 cm - 100 cm*

c) 1 mm - 100 mm *d) 1 m - 100 m*

Ans. : a) 1 nm - 100 nm

vii) *The relation between gross calorific value and net calorific value can be given as* _____.

a) GCV = NCV *b) GCV > NCV*

c) GCV < NCV *d) None of these*

Ans. : b) GCV > NCV

viii) *Acid correction should be* _____ *during calculation of GCV of a fuel by bomb calorimeter correctly.*

a) Added *b) subtracted* *c) multiplied* *d) divided*

Ans. : b) subtracted

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ix) *Which type of reaction occurs in anodic areas ?*

a) Oxidation *b) Reduction*

c) Displacement *d) Addition*

Ans. : a) Oxidation

x) *Poorly oxygenated part becomes _____ whereas well oxygenated part becomes ____ in the differential aeration corrosion.*

a) anode, cathode *b) cathode, anode*

c) anode, anode *d) cathode, cathode*

Ans. : a) anode, cathode

Q.2 a) *What are the causes, disadvantages and preventions of scale formation in the boilers ?* **(Refer section 1.6)** **[6]**

b) *Explain reverse osmosis process for desalination of water with neat labelled diagram.* **(Refer section 1.8.1)** **[3]**

c) *25 ml of water sample required 8.8 ml of 0.04 M EDTA to reach the end point. 25 ml of the same water sample after boiling and filtration required 6.5 ml of same EDTA to reach the end point. Calculate total and permanent and temporary hardness of water sample.* **[3]**

Ans. : Total hardness = 1408 ppm, Permanent hardness = 1040 ppm, Temporary hardness = 368 ppm

OR

Q.3 a) *Explain zeolite process of softening of water with figure, process and reactions.* **(Refer section 1.7.1)** **[6]**

b) *Define hardness of water. Explain the difference between temporary and permanent hardness along with reactions.* **(Refer section 1.3.2)** **[3]**

c) *100 ml of an alkaline water sample requires 5.2 ml of 0.02 M HCl upto phenolphthalein end point and 15.8 ml for methylorange end point. Find the type and amount of alkalinity in water sample.* **[3]**

Ans. : Phenolphthalein alkalinity = 52 ppm, Methylorange alkalinity = 158 ppm

Q.4 a) *Explain pH metric titration of strong acid against strong alkali, with procedure, titration curve and calculations.* **(Refer section 2.11)** **[6]**

b) *Define- i) Specific conductance, ii) Equivalent conductance and iii) Molar conductance.* **(Refer section 2.6.1)** **[3]**

c) *What is reference electrode ? Draw neat labeled diagram of calomel electrode and give its cell representation.* **(Refer sections 2.3 and 2.3.1)** **[3]**

OR

Q.5 a) *Explain different type of electronic transition that occurs in an organic molecule after absorbing UV radiations.* **(Refer section 2.17)** **[6]**

b) *Give the reaction, titration curve and explanation of the conductometric titration of strong acid Vs strong base.* **(Refer section 2.8)** **[3]**

c) *Explain the construction of a conductivity cell with labelled diagram.* **(Refer section 2.7)** **[3]**

Q.6 a) *What is the structural requirement for a polymer to show conductive property ? Give structure, doping and any two applications of polyacetylene.* **(Refer section 3.5.4)** **[6]**

b) *Distinguish between thermosoftning and thermosetting plastics.* **(Refer section 3.4)** **[3]**

c) *Explain the structure of graphene with the help of diagram.* **(Refer section 3.11.1)** **[3]**

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OR

Q.7 a) *What are carbon nano tubes ? Discuss the different types of carbon nanotubes with respect to their structure. Give any two applications of it.* **(Refer section 3.11.2)** **[6]**

b) *Define nanomaterials. How nanomaterials are classified on the basis of dimension, explain any two nanomaterials with examples.* **(Refer sections 3.7 and 3.8)** **[3]**

c) *What is biodegradable polymer ? Explain any two favorable conditions for of biodegradable polymers.* **(Refer section 3.5.3)** **[3]**

Q.8 a) *What is Proximate analysis of coal ? Mention the principle involved in the analysis of each constituent. Give the significance of any one constituent.* **(Refer section 4.5.1)** **[6]**

b) *What is power alcohol ? Give its any two advantages and disadvantages.* **(Refer section 4.6.1)** **[3]**

c) *Calculate % N and % S present in the coal from following data :*
i) 2.1 gm of coal sample in Kjeldahl's experiment liberated ammonia which was absorbed in 40 ml sulfuric acid. The resultant solution required 17 ml of 0.1 N NaOH for complete neutralisation of H_2SO_4 in back titration. The blank titration reading was 28 ml.
ii) 0.55 gm coal was combusted in Bomb calorimeter. Solution from bomb on treatment with $BaCl_2$, formed 0.46 gm $BaSO_4$ dry precipitate. **[3]**

Ans. : % N = 0.733 %, % S = 11.48 %

OR

Q.9 a) *Give construction, working and calculation for finding gross calorific value of a solid fuel by Bomb calorimeter.* **(Refer section 4.3.1)** **[6]**

b) *Explain the production of Hydrogen by steam reforming of methane with reaction conditions.* **(Refer section 4.7)** **[3]**

c) *Observations in the Boy's gas calorimeter experiments are given below. Find GCV and NCV of fuel.*
Volume of gas burnt at STP = 0.08 m^3
Mass of cooling water used = 29.5 kg
Rise in temperature of circulating water = 9.1 °C
Mass of steam condensed = 0.04 kg. **[3]**

Ans. : GCV = 3355.625 kcal/m^3, NCV = 3062.125 kcal/m^3

Q.10 a) *Define corrosion. Explain corrosion by oxygen in case of Na, Cr, Mo, Zn and Au metals with respect to chemical reaction and nature of oxide film formed.* **(Refer section 5.2.1)** **[6]**

b) *Explain the sacrificial anodic protection method for corrosion control. Give any two applications.* **(Refer sections 5.5.1)** **[3]**

c) *What is anodic and cathodic coating ? Which is more protective and why ?* **(Refer section 5.5.3)** **[3]**

OR

Q.11 a) *Define wet corrosion. Explain hydrogen evolution mechanism of wet corrosion with diagram and reactions.* **(Refer section 5.3.2)** **[6]**

b) *Explain any three factors affecting corrosion on the basis of nature of metal.* **(Refer sections 5.4.1)** **[3]**

c) *Draw neat labeled diagram electroplating process with reactions and any two applications.* **(Refer section 5.5.3)** **[3]**

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